

## Using lists

Once a list has been created, you can write programs to manipulate the data inside it—in a loop, for example. You can also combine lists to make new lists.

```
>>> names = ['Simon', 'Kate', 'Vanya']
>>> for item in names:
    print('Hello', item)
```

The list is stored in the variable "names"

The body of the loop must be indented by four spaces

When run, this program displays "Hello", followed by each name on the list

```
Hello Simon
Hello Kate
Hello Vanya
```

```
x = [1, 2, 3, 4]
y = [5, 6, 7, 8]
z = x + y
print(z)
z = [1, 2, 3, 4, 5, 6, 7, 8]
```

Remember, lists are contained within square brackets

This adds the lists together

The new list contains everything from list "x" followed by everything from list "y"



LINGO

## Mutable objects

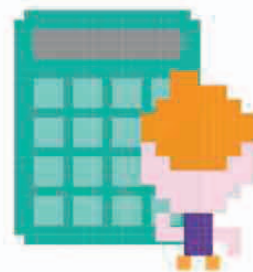
Lists in Python are "mutable". This means that they can change. You can add or delete items, or switch around their order. Other functions in Python, such as tuples (see pp.134–135), can't be altered once you create them. These are called "immutable".

### Lists in loops

You can use a loop to work through every item in a list. This program says "Hello" to a series of names, one after the other.

### Adding lists

Two lists can be added together. The new list will contain the items from both of the old lists.



### Lists in lists

The items in a list can be lists themselves. The "suitcase" list below contains two lists of clothes—it is like a suitcase shared by two people, where they each pack three items.

Because the list is inside square brackets, it becomes an individual item within the "suitcase" list—"suitcase[0]"

"suitcase[1]"

```
>>> suitcase=[['hat', 'tie', 'sock'], ['bag', 'shoe', 'shirt']]
>>> print(suitcase)
[['hat', 'tie', 'sock'], ['bag', 'shoe', 'shirt']]
>>> print(suitcase[1])
['bag', 'shoe', 'shirt']
>>> print(suitcase[1][2])
shirt
```

This will print the whole suitcase list

This will print everything in the second list, "suitcase[1]"

This prints the item at index 2 in "suitcase[1]"—remember, Python starts counting the items from zero

